Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The number of positive integers $< 1,00,000$ which contain exactly one 2, one 5 and one 7 in its decimal representation is

(A) 2940

(B) 7350

(C) 2157

(D) 1582

Q.2

m men and w women are to be seated in a row so that no two women sit together. If $m > w$, then the number of ways in which they can be seated is:

(A) $\frac{m!(m+1)!}{(m-w+1)!}$

(B) ${}^{m}C_{m-w}(m-w)!$

(C) ${}^{m+w}C_{m}(m-w)!$

(D) none of these

Q.3

The number of ways in which we can arrange the digits $1, 2, 3, \ldots, 9$ such that the product of five digits at any of the five consecutive positions is divisible by 7 is

(A) $7!$

(B) ${}^{9}P_{7}$

(C) $8!$

(D) $5(7!)$

Q.4

Let L_1, L_2, L_3 be three distinct parrallel lines in the XY-plane, p distinct the points are taken on each of the three lines. The maximum number of triangles than can be formed by these $3p$ points is:

(A) $p^2(4p-3)$

(B) $p^3 + 3\left({}^{\mathrm{P}}C_2\right)$

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(C) $p^2(3p-4)$

(D) $(p+1)^3-1$

Q.5

The total number of permutations of $n(>1)$ different things taken not more that r at a time, when a thing may be repeated any number of times, is

(A) $\frac{n}{n-1}\left(n^r-1\right)$

(B) $\frac{n^{r-1}}{n-1}$

(C) $\frac{n^r+1}{n+1}$

(D) $\frac{n^r+n}{n-1}$

Q.6

The number of ways of selecting 4 cards of an ordinary pack of playing cards so that exactly 3 of them are of the same denomination is

(A) 2496

(B) ${}^{13}C_3 \times {}^4C_3 \times 48$

(C) ${}^{52}C_3 \times 48$

(D) none of these

Q.7

There are three piles of identical yellow, black and green balls and each pile contains at least 20 balls. The number of ways of selecting 20 balls if the number of black balls to be selected is twice the number of yellow balls, is

(A) 6

(B) 7

(C) 8

(D) 9

Q.8

The number of ways of selecting 4 letters out of the letters of the word MINIMAL is

(A) 16

(B) 17

(C) 18

(D) 20

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Q.9

From 6 different novels and 3 different dictionaries, 4 novels and 1 dictionary are to be selected and arranged in a row on a shelf so that the dictionary is always in the middle. Then the number of such arrangement is

(A) at least 750 but less than 1000

(B) at least 1000

(C) less than 500

(D) at least 500 but less than 750

Q.10

Let $S = \{1, 2, 3, 4\}$. The total number of unordered pairs of disjoint subsets of S is equal to

(A) 25

(B) 34

(C) 42

(D) 41

Q.11

Statement-1: The sum of divisors of $n = 2^{10}3^2 5^3 7^2 11^2$ is
$\frac{1}{480}\left(2^{11}-1\right)\left(3^3-1\right)\left(5^4-1\right)\left(7^3-1\right)\left(11^3-1\right)$
Statement-2: The number of divisors of $m =$
$p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_r^{\alpha_r}$ where $p_1, p_2, \cdots p_{\mathrm{r}}$ are distinct primes
and $\alpha_1, \alpha_2, \cdots \alpha_r$ are natural numbers is $\left(\alpha_1+1\right)\left(\alpha_2+1\right) \cdots \left(\alpha_r+1\right)$.

(A) STATEMENT-1 isTrue, STATEMENT-2 is True; STATEMENT-2 is a correct explanation for STATEMENT-1

(B) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is NOT a correct explanation for STATEMENT-1

(C) STATEMENT-1 is True, STATEMENT-2 is False

(D) STATEMENT-1 is False, STATEMENT-2 is True

Q.12

There are 18 points in a plane such that no three of them are in the same line except five points which are collinear. The number of triangles formed by these points is :

(A) 816

(B) 806

(C) 805

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(D) 813

Q.13

Find the number of non negative solutions of the system of equations: $\mathrm{a}+\mathrm{b}=10, \mathrm{a}+\mathrm{b}+\mathrm{c}+\mathrm{d}=21$, $a+b+c+d+e+f=33, a+b+c+d+e+f+g+h$ $=46$, andso on till $\mathrm{a}+\mathrm{b}+\mathrm{c}+\mathrm{d}+\ldots+\mathrm{x}+\mathrm{y}+\mathrm{z}=208$

(A) ${}^{22}\mathrm{P}_{10}$

(B) ${}^{22}\mathrm{P}_{11}$

(C) ${}^{23}\mathrm{P}_{13}$

(D) None of these

Q.14

Three boys and three girls are to be seated around a table, in a circle. Among them, the boy X does not want any girl neighbour and the girls Y does not want any boy neighbour. The number of such arrangements possible is

(A) 4

(B) 6

(C) 8

(D) None of these

Q.15

Number of ways in which 20 different pearls of two colours can be set alternately on a necklace, there being 10 pearls of each colour.

(A) $6 \times (9!)^2$

(B) $12!$

(C) $4 \times (8!)^2$

(D) $5 \times (9!)^2$

Q.16

A teaparty is arranged for 16 people along two sides of a large table with 8 chairs on each side. Four men want to sit on one particular side and two on the other side. The number of ways in which they can be seated is

(A) $\frac{6!8!10!}{4!6!}$

(B) $\frac{8!8!10!}{4!6!}$

(C) $\frac{8!8!6!}{6!4!}$

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(D) None of these

Q.17

Six teachers and six students have to sit round a circular table such that there is a teacher between any two students. The number of ways in which they can sit is

(A) $6! \times 6!$

(B) $5! \times 6!$

(C) $5! \times 5!$

(D) None of these

Q.18

Six persons A, B, C, D, E and F are to be seated at a circular table. The number of ways, this can be done if A must have either B or C on his right and B must have either C or D on his right is

(A) 36

(B) 12

(C) 24

(D) 18

Q.19

The number of squares that can be formed on a chessooard is

(A) 64

(B) 160

(C) 224

(D) 204

Q.20

Assuming the balls to be identical except for difference in colours, the number of ways in which one or more balls can be selected from 10 white, 9 green and 7 black balls is :

(A) 880

(B) 629

(C) 630

(D) 879

Answers & Solutions

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Q.1 Answer:

2940

Solution:

We may consider a number of up to 5 -digits to be a number of the form XXXXX where X is a digit from 0 to 9. The digit 2 can occupy any of the five places, 3 can occupy any of the remaining 4 places and 7 any of the 3-remaining places. The remaining 2 places can be filled up by 7 digits. Thus, there are $(5)(4)(3)(7)(7)=2940$ positive integers in the desired category.

Q.2 Answer:

$\frac{m!(m+1)!}{(m-w+1)!}$

Solution:

We first arrange the m men.

This can be done in $m!$ ways. After m men have taken their seats, the women must choose w seats out of $(m+1)$ seats marked with X below.

$XMXMXMX \ldots XMX$

1st 2 nd 3 rd $\quad m$ th $(m+1)$ th

They can choose w seats in ${}^{m+1}C_w$ ways and arrange w women in w ! ways.

Thus, the required number of arrangements is

$m!\left({}^{m+1}C_w\right)(w!)=\frac{m!(m+1)!w!}{w!(m+1-w)!}=\frac{m!(m+1)!}{(m+1-w)!}$

Q.3 Answer:

$8!$

Solution:

Let an arrangement of 9 digit number be $x_1x_2x_3x_4x_5x_6x_7x_8x_9$.

Note that we require product of each of

$(x_1,x_2,x_3,x_4,x_5)\,;(x_2,x_3,x_4,x_5,x_6)\,;\ldots;(x_5,x_6,x_7,x_8,x_9)$ is divisible by 7.

This is possible if the 5 th digit is 7.

Therefore, we can arrange the 9 digits in desired number of ways in $8!$ ways.

Q.4 Answer:

$p^2(4p-3)$

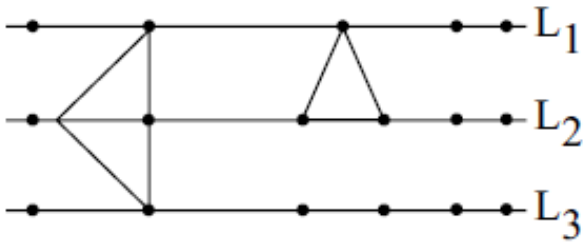
Solution:

To obtain a triangle we may take one point from each of the three lines. This will give us p^3 triangles.

Alternatively, we may choose two lines and take one point on one of them and two points on the other line. This will give us $\left({}^3C_2\right)(2!)(p)\left({}^{\mathrm{P}}C_2\right)=3p^2(p-1)$ triangles

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$\therefore$ **required number of triangles is** $p^3 + 3p^2(p-1) = p^2(4p-3)$



Q.5 Answer:

$\frac{n}{n-1}\left(n^r - 1\right)$

Solution:

When $k(1 \le k \le r)$ **things are arranged, the number of possible arrangements, when repetition is allowed is**

$\underbrace{n \times n \times \ldots\ldots \times n}_{k \text{ times}} = n^k$

Thus, the total possible arrangements is

$n + n^2 + n^3 + \ldots + n^r = \frac{n(n^r-1)}{n-1}$.

Q.6 Answer:

2496

Solution:

We can choose one denomination in ${}^{13}C_1$ **ways, then 3 cards of this denomination can be chosen in** ${}^4C_3$ **ways and one remaining card can be chosen in** ${}^{48}C_1$ **ways. Thus, the total number of choices is** $\left({}^{13}C_1\right)\left({}^4C_3\right){}^{48}C_1) = 13 \times 4 \times 48 = 2496.$

Q.7 Answer:

7

Solution:

Let the number of yellow balls be x**, that of black be** $2x$ **and that of green be** y**. Then** $x + 2x + y = 20$ **or** $3x + y = 20$

$\Rightarrow \quad y = 20 - 3x.$

As $0 \le y \le 20$**, we get** $0 \le 20 - 3x \le 20$

$\Rightarrow \quad 0 \le 3x \le 20 \quad$ **or** $\quad 0 \le x \le 6$

$\therefore \quad$ **The number of ways of selecting the balls is** 7.

Q.8 Answer:

18

Solution:

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Letters of the word, M I N I M A L are (M, $M)(I, I), A, N, L$

We can select 4 letters from letters of the word MINIMAL as follows:

Case 1 **All letters are distinct. This can be done in $^5C_4 = 5$ ways**

Case 2 **Exactly two letters are identical. We can choose two identical letters in $^2C_1$ ways and two remaining distinct letters in $^4C_2 = 6$ ways**

$\therefore$ In this case the selection can be made in $(2)(6) = 12$ ways.

Case 3 **Two pairs of identical letters are selected. This can be done in $^2C_2 = 1$ way**

$\therefore$ two number of ways $= 5 + 12 + 1 = 17.$

Q.9 Answer:

at least 1000

Solution:

We can choose 4 novels out of 6 in $^6C_4$ ways and 1 dictionary out of 3 in $3C_1$ ways. We can arrange 4 novels and 1 dictionary in the middle in $4!$ ways. Thus, required number of ways

$= \left(^6C_4\right)\left(^3C_1\right)(4!) = 1080 > 1000$

Q.10 Answer:

41

Solution:

Let A and B be two subsets of S. If $x \in S$, then x will not belong to $A \cap B$ if x belongs to at most one of A,B.

This can happen in 3 ways. Thus, there are $3^4 = 81$ subsets of S for which $A \cap B = \phi$. Out of these there is just one way for which $A = B = \phi$. As, we, are interested in unordered pairs of disjoint sets, the number of such subsets is

$\frac{1}{2}\left(3^4 - 1\right) + 1 = 41$

Q.11 Answer:

STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is NOT a correct explanation for STATEMENT-1

Solution:

Sum of the divisors of n

$= \left(1 + 2 + \cdots - 2^{10}\right)\left(1 + 3 + 3^2\right)\left(1 + 5 + 5^2 + 5^3\right)$
$\left(1 + 7 + 7^2\right)\left(1 + 11 + 11^2\right)$

$= \left(2^{11} - 1\right)\left(\frac{3^3-1}{3-1}\right)\left(\frac{5^4-1}{5-1}\right)\left(\frac{7^3-1}{7-1}\right)\left(\frac{11^3-1}{11-1}\right)$

$= \frac{1}{480}\left(2^{11} - 1\right)\left(3^3 - 1\right)\left(5^4 - 1\right)\left(7^3 - 1\right)\left(11^3 - 1\right)$

A divisors of m is of the form $p_1^{\beta_1} p_2^{\beta_2} \cdots p_r^{\beta_r}$ where $0 \le \beta_i \le \alpha_i$ for $i = 1, 2, \ldots, r$

That is, β_i can take $\alpha_i + 1$ values. Thus, the number of divisors of m is

$(\alpha_1 + 1)(\alpha_2 + 1)\ldots(\alpha_r + 1)$

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Q.12 Answer:

806

Solution:

A triangle can be formed by using three noncollinear points. So, the number of triangles formed by 18 noncollinear points $= {}^{18}\mathrm{C}_3$

But according to the question, 5 points are collinear. Hence, exact number of triangles $= {}^{18}\mathrm{C}_3 - {}^{5}\mathrm{C}_3 = \frac{18!}{3!15!} - \frac{5!}{3!2!}$

$= \frac{16\times17\times18}{2\times3} - \frac{4\times5}{2} = 816 - 10 = 806.$

Q.13 Answer:

${}^{23}\mathrm{P}_{13}$

Solution:

Consider the equation $\mathrm{a}+\mathrm{b}=10$ **number of solutions of this equation is** ${}^{10+2-1}C_{2-1} = 11$**. Next equation is** $\mathrm{a}+\mathrm{b}+\mathrm{c}+\mathrm{d}=21$ **hence** $\mathrm{c}+\mathrm{d}=11$

and number of solutions of this equation is 12**. Similarly for third equation** $\mathrm{a}+\mathrm{b}+\mathrm{c}+\mathrm{d}+\mathrm{e}+\mathrm{f}$

$= 33$ **or** $\mathrm{e}+\mathrm{f}=12$ **or number of solutions is 13 Similarly for last equation** $\mathrm{a}+\mathrm{b}+\mathrm{c}+\mathrm{d}+\ldots+\mathrm{x}+$

$\mathrm{y}+\mathrm{z}=208$**, or**

$\mathrm{y}+\mathrm{z}=22$ **or number of solution is** 23**. Required number of ways is** $11\times12\times13\times\ldots\times21\times$

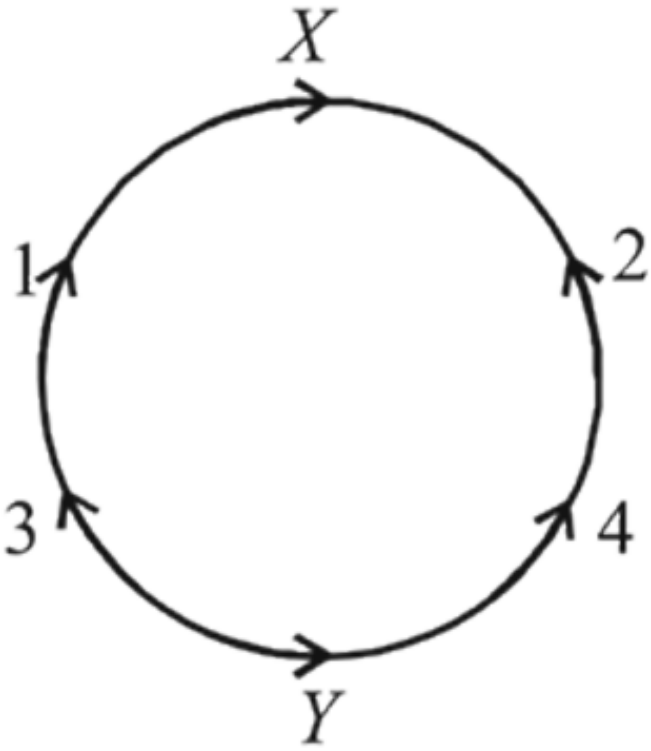
$22\times23 = \frac{23!}{10!} = {}^{23}\mathrm{P}_{13}$

Q.14 Answer:

4

Solution:

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As shown in figure 1,2 and X are the three boys and 3,4 and Y are three girls, Boy X will have neighbours as boys 1 and 2 and the girl Y will have neighbours as girls 3 and 4.
and 2 can be arranged in $P(2,2)$ ways $= 2! = 2 \times 1 = 2$ ways
Also, 3 and 4 can be arranged in $P(2,2)$ ways $= 2! = 2 \times 1 = 2$ ways
Hence, required no. of permutations $= 2 \times 2 = 4$

Q.15 Answer:

$5 \times (9!)^2$

Solution:

Ten pearls of one colour can be arranged in $\frac{1}{2}.(10-1)!$ ways. The number of arrangements of 10 pearls of the other colour in 10 places between the pearls of the first colour $= 10!$
$\therefore$ Required number of ways $= \frac{1}{2} \times 9! \times 10! = 5(9!)^2$

Q.16 Answer:

$\frac{8!8!10!}{4!6!}$

Solution:

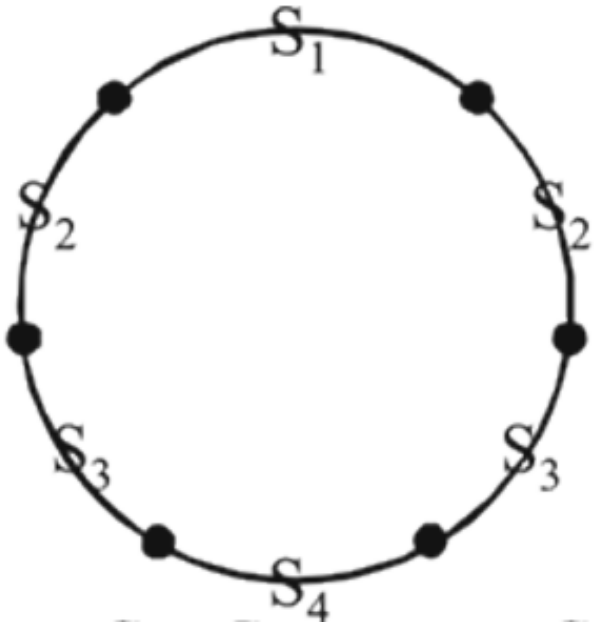
There are 8 chairs on each side of the table. Let the sides be represented by A and B. Let four persons sit on side A, then number of ways of arranging 4 persons on 8 chairs on side $\mathrm{A} = {}^8\mathrm{P}_4$ and then two persons sit on side B. The number of ways of arranging 2 persons on 8 chairs on side $\mathrm{B} = {}^8\mathrm{P}_2$ and the remaining 10 persons can be arranged in remaining 10 chairs in $10!$ ways. Hence, the total number of ways in which the persons can be arranged
$= {}^8\mathrm{P}_4 \times {}^8\mathrm{P}_2 \times 10! = \frac{8!8!10!}{4!6!}$

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Q.17 Answer:

$5! \times 6!$

Solution:



Six students $S_1, S_2, \ldots\ldots, S_6$ can be arranged round a circular table in $5!$ ways. Among these 6 students there are six vactant places, shown by dots $(\bullet)$ in which six teachers can sit in $6!$ ways. Hence, number of arrangement $= 5! \times 6!$

Q.18 Answer:

18

Solution:

When A has B or C to his right we have the order :

AB or AC

When B has C or D to his right, we have the order:

BC or BD Taking these two possibilities together, we must have ABC or ABD or AC and BD. For ABC, D, E, F to arrange along a circle, number of way $= 3! = 6$, where three persons A, B, C together are treated as single. For ABD, C, E, F, the number of ways $= 6$. For AC, BD E, F the number of ways $= 6$ Hence, total number of ways $= 18$.

Q.19 Answer:

204

Solution:

A chessboard is made up of 9 equispaced horizontal and vertical line. To make a 1×1 square, we must choose two consecutive horizontal and vertical lines from among these. This can be done in $8 \times 8 = 8^2$ ways. A 2×2 square needs three consecutive horizontal and vertical lines, and we can do this in $7 \times 7 = 7^2$ ways. Continuing in this manner, the total number of square is $8^2 + 7^2 +$

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$6^2 + \ldots.. + 2^2 + 1^2 = \frac{8(8+1)[(2\times 8)+1]}{6}$
$= 204.$

Q.20 Answer:

879

Solution:

Number of white balls $= 10$ **Number of green balls** $= 9$ **and number of black balls** $= 7$
$\therefore$ Required probability $= (10+1)(9+1)(7+1) - 1$
$= 11.10.8 - 1 = 879$

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